

HLA-Bench: contamination-resistant, executable-oracle evaluation of language models on clinical immunogenetics

DRAFT v0.1 (2026-08-31) — internal working draft for the bioRxiv preprint. Author: Jason Brelsford (Brelsford Software LLC). Numbers below are from the committed results in `bench/`; every figure will be regenerated with provenance before submission.

Licence: this manuscript will be submitted under CC-BY 4.0. Reference data (IPD-IMGT/HLA) is CC-BY-ND and is cited and computed upon, never redistributed; no IPD-IMGT/HLA-derived data files will be uploaded as supplementary material — see §6 and `docs/PREPRINT_CHECKLIST.md`.

Abstract (draft)

Large language models are entering clinical genomics workflows, but no benchmark measures whether they handle the field's foundational layer: HLA nomenclature and donor–recipient histocompatibility assessment, where a fabricated allele name or a miscounted match is a patient-safety error. We present HLA-Bench, two evaluation suites — 550 nomenclature tasks (family A) and 205 donor–recipient matching pairs (family C) — in which every answer is computed by an executable oracle from the pinned IPD-IMGT/HLA release itself: no human labels, no LLM judges, no licensed data. Because ground truth regenerates deterministically from each quarterly database release, a versioned share of tasks is post-training-cutoff by construction, giving contamination resistance that static benchmarks cannot offer. Across eight open-weight models (3B–14B) and one frontier model, accuracy on nomenclature ranged from 15% to 45.5% against a 28% naive string-manipulation baseline, and **every model family tested scored 0% on two-field ambiguity expansion** (0 of 30 tasks for each of the nine models) — the core clinical trap that a two-field name denotes many full-resolution alleles. On matching, string manipulation collapses to 0% and the best tested open model reached 13.7%; the dominant errors are counting matched loci instead of chromosomes, inventing the denominator, and over-crediting matches, and no model ever flagged unresolvable typing. Models fabricated allele names at 0.06–0.20 per task on the full 550-task suite; claude-sonnet-4-6, the most accurate model tested, fabricated at 0.11 per task and was calibrated on only 77% of its answers. A GRPO fine-tune on a disjoint generated split [RESULTS PENDING] demonstrates the suites function as training environments, not only evaluations. All generators, graders, and the verification service are open source; the scored split remains sealed and regenerates every release.

1. Introduction

- AI assistance is reaching clinical genomics: agentic "AI workbench" products now ship genomics database connectors and multi-agent pipelines. Their built-in reviewers check citations and arithmetic — none holds allele-level ground truth.
- HLA is uniquely suited to deterministic evaluation: the field's single authoritative, versioned, machine-readable source of truth (IPD-IMGT/HLA; 46,652 named alleles at release 3.65.0; 288 deleted names; quarterly releases) makes correctness computable rather than adjudicated.
- The stakes are concrete: a nomenclature error propagates into registry searches and match assessments; the classic traps (null alleles inside apparent serologic matches; legacy-era typings; resolution-insufficient reports read as confident counts) are exactly where clerical-style AI errors become clinical ones.
- Contributions: (i) two executable-oracle suites with tiered difficulty and audited failure taxonomies; (ii) a contamination-resistance design that keys ground truth to database releases; (iii) results for seven models with stratified wrong-answer audits; (iv) an open verification service (HLA-Verify) exposing the graders as a runtime guard; (v) external validation: 99.3% concordance with NMDP's py-ard on the overlapping reduction slice, with all divergences triaged; (vi) [pending] a training-environment demonstration via GRPO on a disjoint split.

2. Related work

- **LLM genomics benchmarks.** GeneTuring: 16 tasks / 1,600 curated questions, 48,000 answers evaluated manually (Shang et al., Brief Bioinform 2025, doi:10.1093/bib/bbaf492) — curated QA with static keys, no immunogenetics. LAB-Bench (Laurent et al., arXiv:2407.10362) and BixBench (arXiv:2503.00096) cover biology research and computational-biology agent tasks. GAME (doi:10.1101/2025.07.04.663250) standardizes benchmarking of genomic sequence-to-activity models — a different model class.
- **LLMs vs deterministic tools.** Reese et al. (doi:10.1101/2024.07.22.24310816) benchmark seven LLMs against Exomiser on 5,213 rare-disease cases: best LLM 23.6% top-1 vs 35.5% for the deterministic tool — converging evidence that LLMs trail executable domain logic; our design makes the executable logic the grader rather than the competitor.
- **Public-data validation in HLA methodology.** Multi-ethnic HLA imputation reference panels are routinely built and validated on 1000 Genomes-class open data (Degenhardt et al., doi:10.1093/hmg/ddy443; Silva et al., doi:10.1111/tan.15543; Barbosa et al., doi:10.1016/j.humimm.2026.111700). We follow this established public-data pattern and extend it from imputation models to language-model evaluation.
- **HLA reduction tooling.** py-ard (NMDP's ARD-reduction library) is the field-standard normalizer for *valid* typing inside matching pipelines. Our concordance study (docs/pyard-concordance.md; 2,000-allele deterministic sample) finds 99.3% agreement on the overlap slice; all divergences fall into two documented semantic families (group-vs-allele suffix handling; ARD rollups). Critically, on AI-generated input the two tools answer different questions: given the fabricated `DQB1*05:03:26:99`, py-ard reduces it to a plausible `DQB1*05:03`, while a verifier must instead classify it as fabricated — reduction and verification are different jobs, and evaluating models requires the latter.
- **Gap.** No prior benchmark combines clinical immunogenetics content, LLM/agent evaluation, executable-oracle grading, and release-keyed regeneration of ground truth.

3. Benchmark design

3.1 Executable oracle

Every task's answer is computed from the pinned release's own files (Allelelist, Deleted_alleles, WMDA rel_dna_ser / rel_ser_ser, nomenclature history) by rules documented for audit (normalization rules; matching rules R1–R6). The oracle scores 100% by construction; a grader change is only ever made with a failing test written first. No human annotation, no LLM-as-judge.

3.2 Contamination resistance

Suites regenerate deterministically from (release tag, seed). 32.7% of tier-3/4 allele tasks in family A involve database facts newer than an assumed model training cutoff (release 3.58.0); this share refreshes every quarterly release. Task identifiers embed a suite revision; sealed scored splits (438/550 and 162/205 tasks) never leave the grading machines, while dev splits are public.

3.3 Family A — nomenclature (550 tasks, 20 subtypes, 4 tiers)

Truncation with expression-suffix rules, validity, G/P group assignment, serologic equivalents, rename/deletion history, ambiguity expansion, typing-report normalization, near-miss discrimination, release drift. Object answers graded exactly; every emitted allele-shaped token checked against the release (fabrication detection); confidence and flags graded for calibration.

3.4 Family C — donor–recipient matching (205 pairs, 8 subtypes)

Frameworks 6/6 through 12/12 and antigen-level; mixed typing eras and resolutions. Rules R1–R6 encode: normalize-then-compare (deleted names chased to successors), per-chromosome multiset counting, WMDA-column antigen assignment with null handling ('o' and suffix N express no antigen), GvH/HvG directionality, and the R6 principle that unresolvable typing yields a potential verdict excluded from the denominator — a confident count over unresolvable typing is itself the graded error.

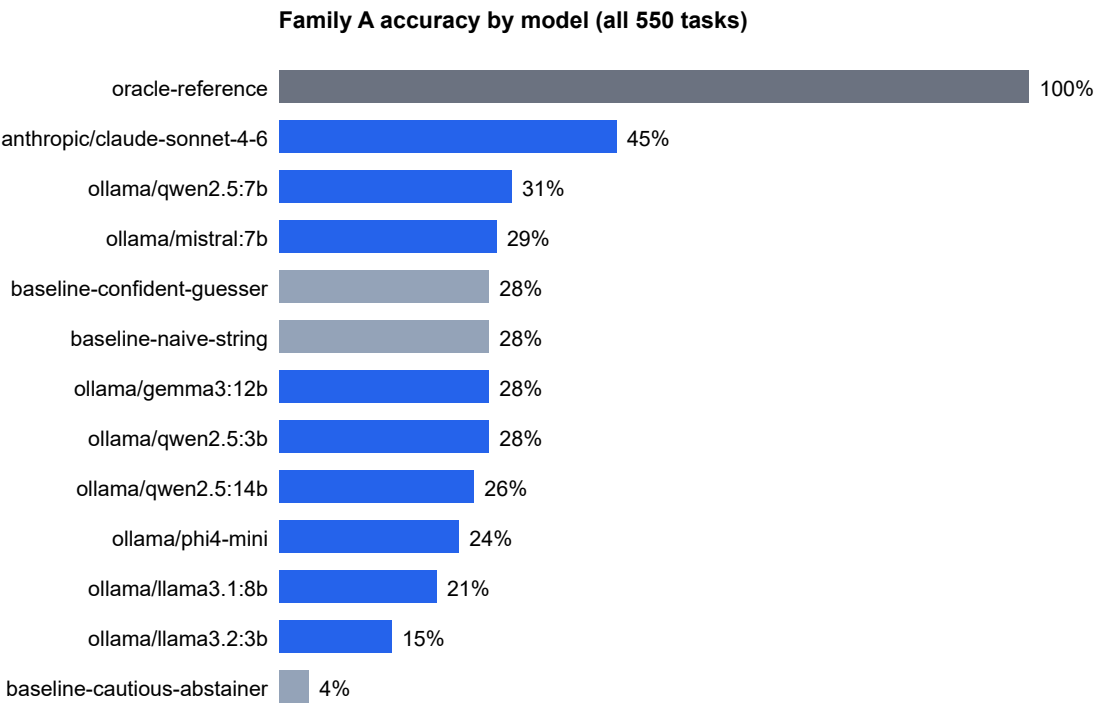
3.5 Harness

Local open-weight models via Ollama (temperature 0, JSON-forced); hosted models via vendor APIs with a bare-JSON clamp (strict system prompt; Anthropic prefilled-`{` continuation; OpenAI `json_object`; Gemini JSON MIME) recorded as `prompt_rev` in results. Wilson 95% CIs throughout. A stratified wrong-answer sample (≤ 3 per subtype) is committed per run so failure shapes are auditable without exposing the sealed split.

4. Results

4.1 Family A (all 550)

Figure 1. Accuracy by model, all 550 tasks. Generated deterministically from `bench/HLA-Bench-A.md`'s headline table by `scripts/generate_figures.py`; regenerate after any results update. Gray = oracle (validates the harness, not a model result); slate = non-LLM baselines; blue = evaluated models.



Family A accuracy by model, all 550 tasks

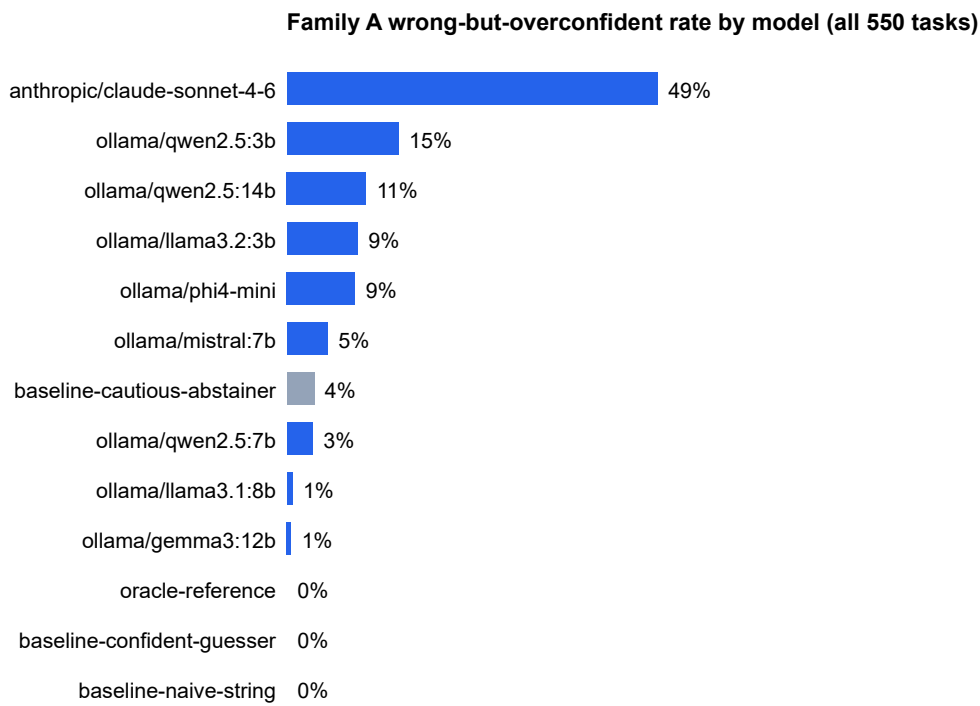
Model	Acc [95% CI]	Fabricated- name tasks	Notes
oracle	100%	0	validates harness
claude-sonnet-4-6	45.5% [41.3–49.6]	54	most accurate model tested; 0.11/task fabrication; calibrated on 77% of answers; <code>wrong_calibrated</code> is the most common outcome (272/550)
qwen2.5:7b	31% [27–35]	47	perfect <code>null_trap</code> ; 0% <code>expand_ambiguity</code>
mistral:7b	29% [26–33]	76	truncation changes digits (<i>B15:504</i> → <i>B15:01</i>)
naive-string baseline	28%	0	
qwen2.5:3b	28% [24–32]	27	most cautious; best calibration
gemma3:12b	28% [25–32]	80	highest fabrication rate (0.20/task); 93% <code>locus_field</code> , 87% <code>truncate</code> ; 0% <code>expand_ambiguity</code> , 0% <code>null_trap</code>
qwen2.5:14b	26% [23–30]	39	refuses 120/550 (22%); 33% on answered; 90% <code>null_trap</code> , 80% <code>near_miss</code> ; 0% <code>expand_ambiguity</code>
phi4-mini	24% [21–28]	35	
llama3.1:8b	21% [18–25]	43	worst calibration (402/550 wrong-overconfident)
llama3.2:3b	15% [12–18]	57	below the string baseline
cautious-abstainer	4%	0	refusal floor

Fabrication rate per task, full suite: 0.06 (qwen2.5:3b) to 0.20 (gemma3:12b).

Key findings: (1) universal 0% on `expand_ambiguity` across Claude, Qwen, Mistral, Llama, Phi, and Gemma, at every scale from 3B to 14B — no tested model knows a 2-field name covers 2–389 full-resolution alleles; (2) a 3B model scores below string manipulation, and so does a 14B model: qwen2.5:14b refuses 22% of tasks outright and, on the tasks it answers, matches rather than beats its 7B sibling (33% vs 31%) — added parameters buy caution on the traps it recognises (null alleles, near-misses) but no nomenclature knowledge; (3) fabrication is universal and takes characteristic forms (invented 4th fields, legacy colon-less strings, invented G/P group names); (4) reasoning-mode models under forced-JSON burn their entire budget thinking (deepseek-r1:8b, excluded; harness finding); (5) a second harness finding: a local inference server sized to a

model's full training context (131k for gemma3:12b) crashed repeatedly on consumer hardware and returned empty bodies with HTTP 200 for 460/550 requests — the harness now pins the context window, retries empty replies, and refuses to cache them, and the reported Gemma row is the clean re-run.

Figure 2. wrong_but_overconfident rate by model — the share of tasks where a model states a factually wrong allele name without flagging uncertainty, the failure mode with the most direct clinical cost. Generated deterministically from bench/HLA-Bench-A.md's failure-modes table by the same script as Figure 1; same coloring.

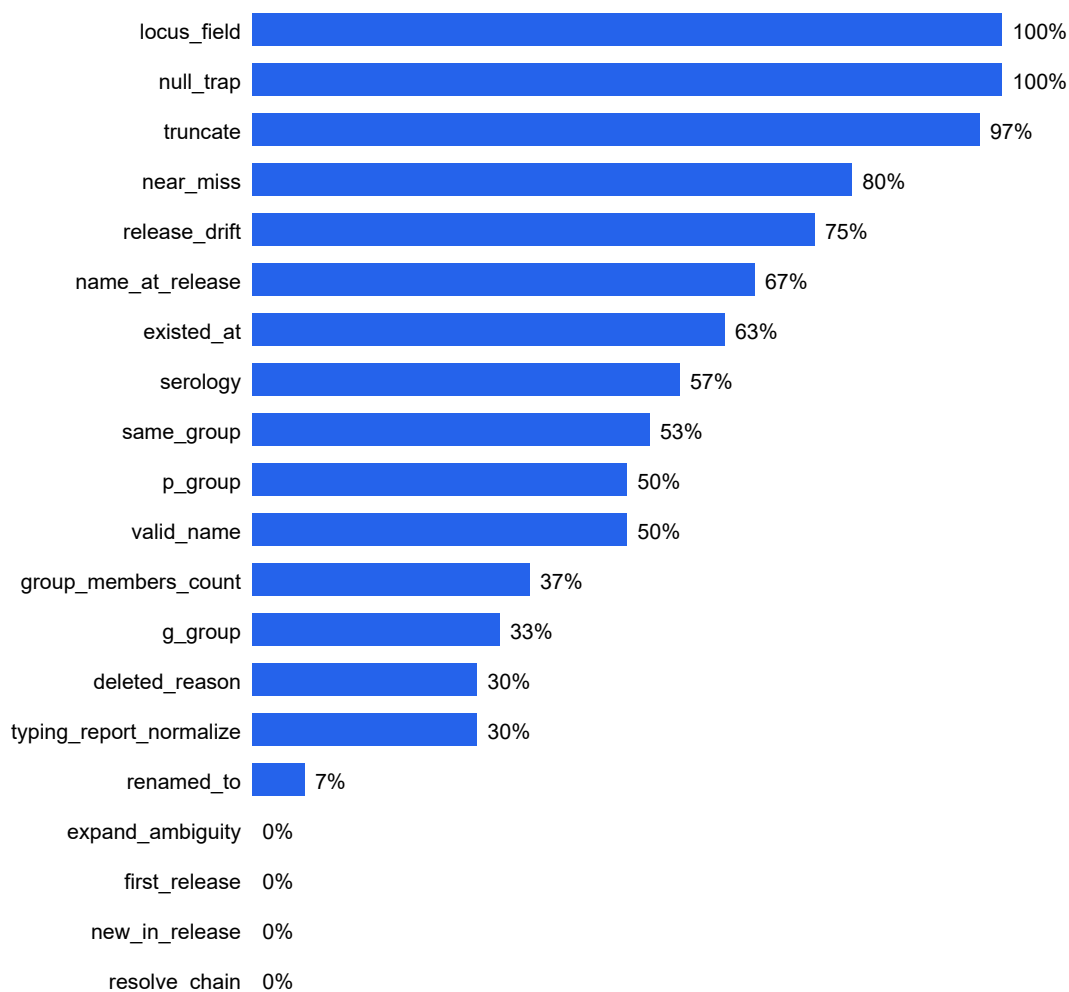


Family A wrong-but-overconfident rate by model, all 550 tasks

The ranking does not track model size or accuracy: llama3.1:8b is both a mid-table accuracy model (21%) and the worst on this metric (73%), while claude-sonnet-4-6 is lowest (4%) even in the clean, fully-answered re-run — its dominant failure mode is wrong_calibrated (272/550), not confident wrongness. Both non-LLM baselines sit at 53%, which is the number a tested model needs to beat to demonstrate real calibration rather than confident guessing; qwen2.5:14b (40%), qwen2.5:3b (45%), phi4-mini (46%), and claude-sonnet-4-6 (4%) do, the other five tested models do not.

Figure 3. Per-subtype accuracy for claude-sonnet-4-6, the most accurate model tested, from bench/HLA-Bench-A.md's by-subtype table (all split). Same script and coloring as Figures 1-2.

Family A accuracy by subtype (anthropic/claude-sonnet-4-6, all 550 tasks)



Family A accuracy by subtype, anthropic/claude-sonnet-4-6, all 550 tasks

The best tested model is perfect on `locus_field` and `null_trap` (both 100%) and near-perfect on `truncate` (97%), but four subtypes sit at 0%: `expand_ambiguity`, `first_release`, `new_in_release`, and `resolve_chain`. Three of those four require reasoning across the full release history (when a name first appeared, what changed it, what a renamed/deleted chain resolves to) rather than parsing a single name in isolation, and `expand_ambiguity` repeats the universal failure already noted in Figure 1's discussion — no model at any scale tested knows how many full-resolution alleles a 2-field name covers. Accuracy is not a single number per model; it is a profile of which lookups a release-pinned table answers and which a model still has to guess at.

4.2 Family C (all 205)

Model	Acc [95% CI]
oracle	100%
mistral:7b	13.7% [9.6–19.0]
qwen2.5:14b	11.7% [8.0–16.8]
qwen2.5:7b	6.3% [3.7–10.6]
llama3.1:8b	3.9% [2.0–7.5]
gemma3:12b	0% [0–1.8]
phi4-mini	0% [0–1.8]
naive-string baseline	0% [0–1.8]
cautious-abstainer	0% [0–1.8]

Unlike nomenclature, matching admits no string shortcut: both baselines collapse to zero. The best open model, mistral:7b (13.7%), keeps the denominator right but over-credits matches ("8/8" for a 6/8 pair) — a missed mismatch, the clinically more dangerous direction — and is 0% on resolution-insufficient typing. Doubling model size (qwen2.5:14b, 11.7%) nearly doubles the score, but the whole gain sits on the simplest counting subtype (60% vs 33%); the 14B model still returns "4/4" for an 8/8 framework and "5/6" for 10/10 — the wrong denominator as well as loci-not-chromosomes — and is 0% on every clinical-risk slice, with 181/205 answers wrong-but-overconfident and no `potential` verdict ever emitted. llama3.1:8b (3.9%, 197/205 wrong-but-overconfident) invents denominators the same way ("6/6" for an 8/8 framework). phi4-mini scores 0/205, 139 of them schema failures (a bare count or "match" in place of the per-locus object, sometimes with the right number) — an unparseable verdict is graded as no verdict. gemma3:12b scores 0/205 (203 wrong-but-overconfident): it reports the locus count as both numerator and denominator — "4/4" for every 8/8 pair — so it cannot score even a perfect match correctly. qwen2.5:7b succeeds only on tier-1 counting (22%); 0% on antigen-vs- allele, null traps, directionality, framework shifts, and resolution- insufficient cases. Hand-audit of the stratified wrong sample shows two dominant error modes: counting matched **loci** instead of chromosomes ("4/8" when four loci match), and self-contradiction (all-match per-locus verdicts beside a 5/10 count). The model never emitted a `potential` verdict — the clinically dangerous confident-count behaviour the suite was built to expose. 190/205 answers were wrong-but-overconfident.

4.3 Training-environment demonstration [PENDING]

GRPO fine-tune of a 3B model on a disjoint generated split (separate seed, 0 task-id overlap), evaluated on the sealed split before/after. [Table: base vs tuned accuracy, fabrication rate, calibration. Run planned on a rented A100; scripts in `docs/` `runbook`.]

4.4 Reproducibility across environments

The family-C suite regraded to an identical score (n=205, 13/205 correct, identical CIs, identical per-subtype and per-tier tables) across three runner configurations: the original interactive runner; a torn-down-and-reinstalled service runner; and a third configuration under a different Windows service account, Python 3.14 instead of 3.12, and a different launcher path — same (release tag, seed) each time, consistent with the deterministic-seeding design (sha256-derived per-task seeds). Across the three runs, exactly one of the 205 raw model responses differed in form (a malformed JSON object in place of a wrong-but-overconfident answer at temperature 0 — inference-backend nondeterminism, not the grader); it was scored wrong in both forms, so the score and every reported table were unaffected. We report this rather than claim raw-response determinism, which no local inference stack guarantees.

4.5 External validation against NMDP tooling

On a deterministic 2,000-allele sample across six loci, our normalizer's 2-field reductions agree with py-ard 2.4.2 (db 3650) on 1,986/2,000 (99.30%). The 14 divergences comprise seven Q-suffix cases (we name the 2-field group; py-ard annotates the reported allele's own suffix) and seven ARD-equivalence rollups (py-ard maps ARD-identical alleles to a group exemplar — correct for matching, lossy for name verification). Full table and triage in the repository (`docs/pyard-concordance.md`), regenerated by CI from `scripts/pyard_concordance.py`.

5. Discussion

- Deterministic verification is complementary infrastructure for agentic science tools: the graders run as a service (HLA-Verify) gating model output at runtime, the same rules that grade the benchmark.
- Limitations: English-only prompts; vo scope excludes DPB1 TCE permissiveness, DQA1/DPA1, and frequency-weighted population realism (data hierarchy in available on request); single-turn tasks (agentic multi-step variants are future work).
- Contamination resistance is structural, not assumed: post-cutoff facts did not yield the expected advantage-from-memorization pattern (post-cutoff slice scored *higher* than deleted-name slice for qwen2.5:7b, consistent with task- type mix rather than memorization; per-subtype breakdown in supplement).
- The 0% ambiguity-expansion result across all families suggests a shared representational gap, not a scale problem (identical failure at 3B and 7B and in the frontier model), making it a concrete target for environment-driven training.
- Across all tested models, `claude-sonnet-4-6` is now clearly the most accurate on Family A, but higher accuracy does not close the safety gaps: it still scores 0% on `expand_ambiguity`, still fabricates allele names at 0.11 per task, and is still calibrated on only 77% of its answers.

6. Data and code availability

Generators, graders, harness, adapters (verifiers/Inspect AI), and the verification service: github.com/jasonbrelsford/verifiable-science-envs (Apache-2.0; service under PolyForm-NC). Reference data fetched at runtime from IPD-IMGT/HLA (CC-BY-ND; Barker DJ et al., NAR 2025) and never redistributed. Dev splits public; sealed splits regenerate per release from (tag, seed). In-browser demo: hlaverify.com/demo.

7. Competing interests

J.B. is the founder of Brelsford Software LLC, which operates the HLA-Verify API (hlaverify.com) described in this manuscript as a commercial service under the PolyForm Noncommercial 1.0.0 licence for the verification component; the benchmark, generators, and graders themselves are released under Apache-2.0.

TODO before submission

- [x] Clean claude-sonnet-4-6 re-run at 1600 tokens (2026-09-28, commit 5248aec; 45.5% [41.3-49.6], 250/550, all tasks answered)
- [] GRPO delta table (Modal A100, ~\$10–30, likely within free credits)
- [x] 14B tier row (qwen2.5:14b, 2026-09-09; further 12–14B families queued on TOWER)
- [] Per-subtype post-cutoff breakdown (contamination supplement)
- [] Cross-machine bit-identical reproduction check (tower vs tower2)
- [x] Figure 1: Family A accuracy by model (figures/family-a-accuracy-by-model.svg, scripts/generate_figures.py)
- [x] Figure 2: Family A wrong-but-overconfident rate by model (figures/family-a-wrong-but-overconfident-by-model.svg, same script); fixed a footnote-marker bug in the parser while adding it, so Figure 1 now includes the previously-dropped claude-sonnet-4-6 row too
- [x] Figure 3: Family A per-subtype accuracy for the best tested model (figures/family-a-accuracy-by-subtype.svg, same script, new parse_by_subtype)
- [] Additional figures with provenance (calibration); bioRxiv category: bioinformatics
- [] Decide author list / acknowledgements; ORCID